

30563 – MATHEMATICAL MODELLING FOR NEUROSCIENCE
Final Project Report

The Shape of a Stimulus

*Study of geometry of three stimulus classes
in mouse visual cortex (MICrONS)*

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1 Introduction and Research Question

A stimulus, to visual cortex, is not a property of any single neuron. It is a pattern: thousands of cells firing together, settling into a particular region of activity space. The geometry of that region, where it sits, how it drifts, how far it stands from other stimuli, is what carries class identity [2].

We can use two tools to probe this geometry from opposite ends. PCA looks for the axes that explain the most variance and ignores any labels we might have. A supervised decoder does the opposite: it goes straight for the directions that pull the classes apart. When the two methods disagree, the disagreement is the finding. It says the class signal lives in low-variance directions that PCA was never asked to find.

The MICrONS Functional Connectomics dataset [1] lets us run that comparison at scale. A single session records around 8,000 neurons across primary visual cortex (V1) and three of its dorsal-stream targets (AL, LM, RL) while the mouse watches three stimulus classes spanning the natural-to-synthetic range: Clip (natural movies), Monet2 (textured noise), and Trippy (synthetic waves). The question that started the project was whether the three classes occupy distinguishable population states, and whether that separation holds equally across all four areas. A first round of results pushed us to a sharper one: is the three-way structure real, or a one-dimensional naturalness boundary in disguise? And if it is a boundary, does any one area specialise on it per neuron?

2 Methodology

Session selection. The fourteen functional sessions in MICrONS were screened on five summary statistics (neuron count, median per-neuron activity, mean pupil diameter, mean treadmill speed, and neuron distribution across brain areas). Anything outside $|z| > 2$ was flagged as outlier (Fig. 1). Session 7_5 sat at the median across all metrics: 8,194 neurons against a cohort median of 8,176, no behavioural red flags, and a representative profile across all the sessions.

Preprocessing. Eleven trials with running speed $|v| > 1$ were dropped. Although 2.4% of the session, there are enough points to influence the response of the visual cortex. Each neuron’s full timeseries was then linearly detrended (removing the $\sim 45\%$ photobleaching drift identified earlier) and z-scored. 453 trials remained: 377 Clip, 38 Monet2, 38 Trippy. The trial-averaged population response after this pipeline already separates the three stimulus classes by mean activity in every cortical area (Fig. 2).

Three experiments. Experiment 1 collapses each trial to a single mean activity vector, fits PCA per area, and quantifies separability in two ways: balanced silhouette in the top three PCs, and 5-fold cross-validated multinomial logistic regression on the full neuron matrix (supervised). Experiment 2 keeps the time axis. For each area and class it builds a trial-averaged $75 \times N$ trajectory and measures pairwise Euclidean distance frame by frame in two spaces: the top-3 PC subspace (the picture you see in figures) and the full neuron space (the test we trust). Significance is assessed by 100 label-permutation shuffles and variability by 1,000 class-stratified bootstrap resamples, with the PCA basis held fixed throughout, so what the envelopes and nulls capture is uncertainty in the trajectories themselves, not in the projection. Experiment 3 runs the same pipeline of Experiment 2 inside the Clip category, using the three content sub-categories MICrONS provides (Cinematic, sports1m, Rendered).

Checks. Two confounds were obvious. Clip has roughly ten times the trials of both synthetic classes, so we sub-sampled it down to 38 and recomputed. V1 has more than thirteen times AL’s neuron count, so every area was sub-sampled to AL’s 414 neurons (20 random draws) and the entire pipeline was rerun.

3 Results

3.1 A signal exists but PCA is not aligned with it (Exp. 1). Each trial reduces to a single N -dimensional vector whose i -th component is neuron i ’s mean response over the trial. Each trial becomes a point, with one coordinate per neuron, so the whole session is a cloud of these points. The silhouette score and the classifier both work on this exact same cloud of points. Cross-validated logistic regression on it reaches 98.9% accuracy in V1, 98.0% in LM, 97.8% in RL and 94.7% in AL, well above the 83.2% majority baseline. Viewed through the top three principal components of that same cloud the picture changes: balanced silhouette oscillates around zero in every area (Fig. 3). The class signal is there, it is indeed decodable, but it does not live in the directions PCA prioritises. A standard PC-space scatter

would have shown nothing (Fig. 4), which is what pushed us toward the time-resolved trajectory analysis that follows.

3.2 Trajectories: an artefact, then the real answer (Exp. 2). Now we retain the time axis. For each area and class, same-class trials are averaged frame by frame to give a trajectory: 75 successive points in neuron space, one per frame, each an N -dimensional vector holding the across-trial mean activity at that timepoint. Monet2 and Trippy trials had originally 113 frames but we truncated them to Clip’s 75 so the three trajectories are directly comparable. By plotting the trajectories in the PC1-PC2 plane, every area produces three clean, stationary and non-overlapping clusters (Fig. 5). The 75 frames cluster in one spot instead of moving along a path, so each class sits at one fixed location in state space. A first significance test, run in that same top-3 PC space, returned every pair separated at $p \approx 0.01$. But this test is not what it appears. The PCA basis was fit on real-label trial averages, and the shuffled trajectories were then projected through the same basis, so the projection itself carries label information and the shuffle null can never reach what the real data reach. Therefore we re-run the test in the full neuron space, where there is no projection, and the results are completely different (Fig. 6). Table 1 reports the honest-test p -values. Monet2 $\leftrightarrow$ Trippy does not separate anywhere. The only synthetic pair that survives is Clip $\leftrightarrow$ Trippy in RL ($p = 0.02$). The Clip-vs-synthetic distinction, on the other hand, is consistent across every area. Both controls behave as expected: Clip sub-sampled to 38 trials reproduces the same curves, and matching every area to AL’s neuron count collapses V1’s apparent advantage, leaving the four areas in a tight per-neuron band (Fig. 7).

3.3 Inside Clip, AL’s advantage tracks naturalness (Exp. 3). By applying the same trajectory pipeline to Clip’s three content sub-categories, every pair separates at $p \leq 0.02$ in every area, becoming significant by frame 2 (~ 267 ms post stimulus onset, Fig. 9). Moreover, after population matching, AL outperforms every other area on the two pairs that cross the natural/synthetic boundary (Cinematic $\leftrightarrow$ Rendered and sports1m $\leftrightarrow$ Rendered), but matches the others on Cinematic $\leftrightarrow$ sports1m, which is natural-vs-natural (Fig. 10). The simplest interpretation is that AL is not encoding content category at all. It is encoding the same naturalness axis Experiment 2 showed, now recurring within Clip itself.

4 Discussion

The Clip-vs-synthetic boundary held up under every control we ran and showed up in every area, so cortex does carry that signal linearly, and it does so uniformly across the visual hierarchy. The Monet2 $\leftrightarrow$ Trippy boundary is a different story. It is invisible to linear methods at the population scale. PCA can be made to display it, but only because the basis itself was informed by the labels. What both experiments end up pointing at is the same organising dimension: naturalness explains the structure between classes in Experiment 2, and reappears in Experiment 3 as the axis along which AL gains a per-neuron edge. Taken together, this suggests that what differs across areas is sensitivity to natural-vs-artificial structure, not sensitivity to content category.

Fitting PCA on real labels and then running shuffled data through the same projection is a self-confirming test. Once the projection step has seen the labels, the null is compromised, and significance has to be assessed in the unprojected space. We learned this the hard way in Experiment 2, and Experiment 3 was built around the fix from the start.

Four limitations qualify the biological reading. *(i)* We cannot cleanly separate category encoding from low-level image statistics like brightness, contrast and motion energy, since the dataset does not provide stimuli matched on those statistics. *(ii)* Everything here comes from a single session, so cross-mouse generalisation is untested. *(iii)* Linear methods may simply be missing non-linear structure, and the Monet2 $\leftrightarrow$ Trippy collapse could be a property of the model rather than of the cortex. *(iv)* We did not use the connectomics half of MICrONS, so the structural-to-functional link is still open.

Three things would push the project forward. First, a non-linear method such as CEBRA [3] aimed at the Monet2 $\leftrightarrow$ Trippy contrast, to see whether it finds what PCA misses. Second, cross-session replication of AL’s apparent role as a naturalness detector, to check it is not a peculiarity of session 7_5. Third, a test of whether AL’s per-neuron advantage aligns with the wiring patterns visible in the EM volume, the question MICrONS was designed to make tractable.

References

- [1] The MICrONS Consortium. *Functional connectomics spanning multiple areas of mouse visual cortex*. Nature, 2025.
- [2] P. Dayan and L. F. Abbott. *Theoretical Neuroscience*. MIT Press, 2001.
- [3] S. Schneider, J. H. Lee, M. W. Mathis. *Learnable latent embeddings for joint behavioural and neural analysis*. Nature, 617, 360–368 (2023).

A Tables

Area	Clip↔Monet2	Clip↔Trippy	Monet2↔Trippy
V1	0.58	0.06	0.37
AL	0.45	0.22	0.76
LM	0.29	0.09	0.48
RL	0.10	0.02	0.29

Table 1: Honest-test p -values: maximum full-feature trajectory distance against the 100-shuffle null. Monet2↔Trippy never crosses the threshold in any area.

B Figures

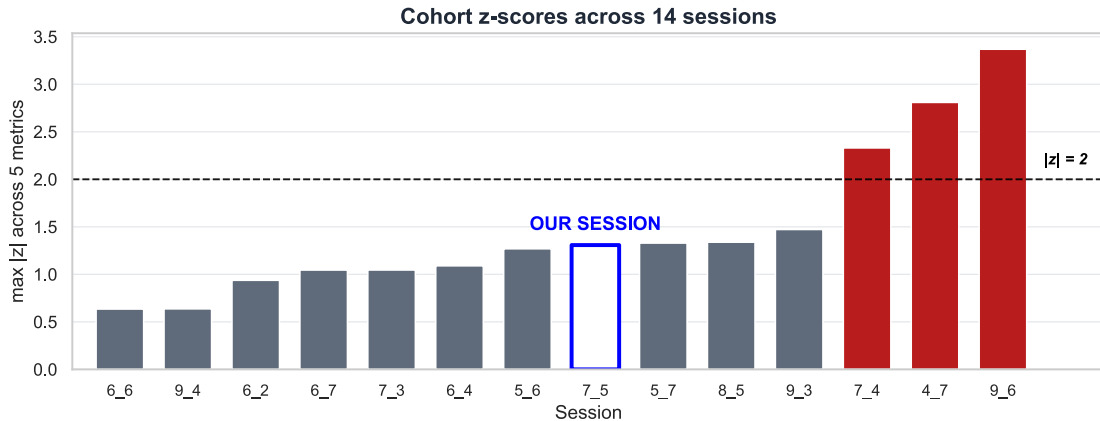


Figure 1: **Cohort screening across the 14 MICrONS sessions.** Per-session maximum $|z|$ across five summary statistics (neuron count, median per-neuron activity, mean pupil diameter, mean treadmill speed, fraction of unknown trials). Coral bars cross the $|z| = 2$ flag; session 7_5 (white bar) sits comfortably below the threshold on every axis and was selected for the deep dive.

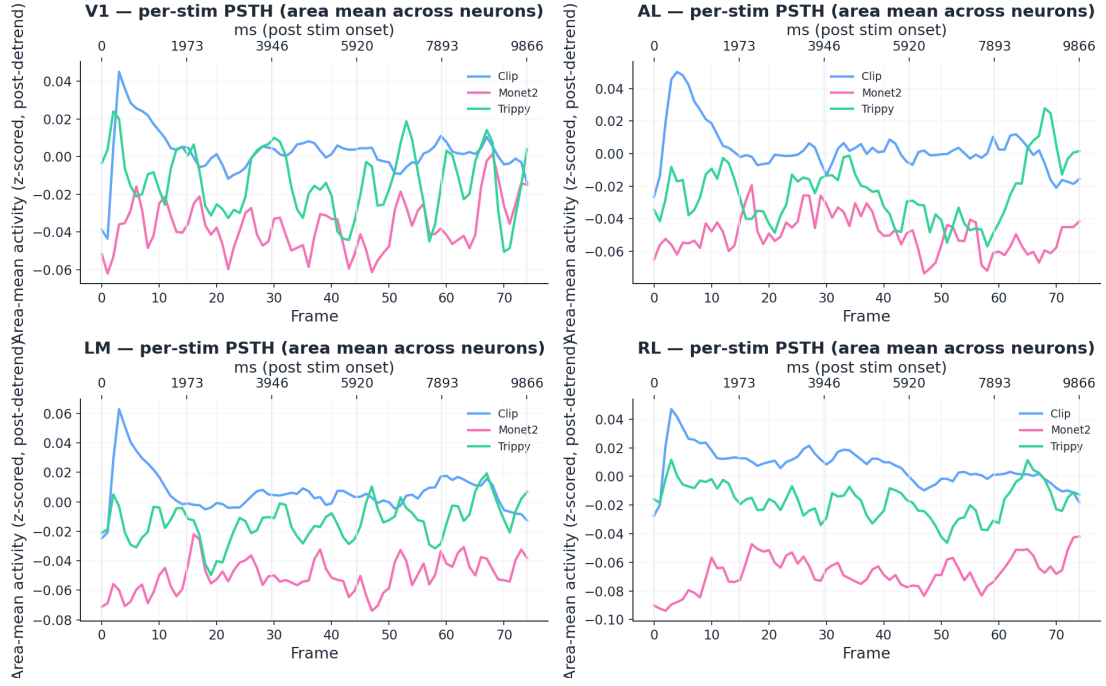


Figure 2: **Trial-averaged PSTH per stimulus class, per cortical area.** Mean population response averaged within each stimulus class and plotted against trial time, one panel per area (V1, AL, LM, RL). Trippy sits highest, Clip intermediate (with an onset transient in the first few frames), Monet2 lowest. Class identity is already encoded in the first moment of population activity, before any dimensionality reduction.

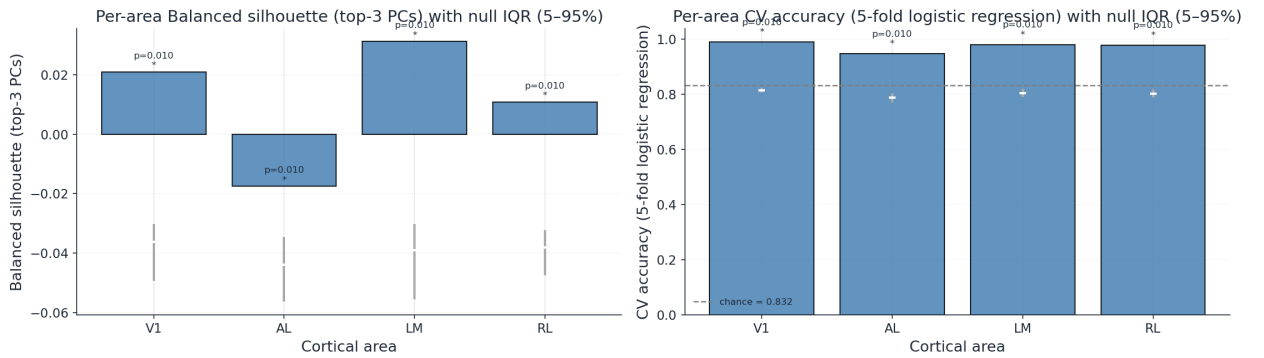


Figure 3: **Two separability metrics across cortical areas (Exp. 1).** Left: balanced silhouette in the top three PCs, with per-area shuffle-null IQR overlaid as a grey vertical bar. Right: 5-fold cross-validated multinomial logistic regression accuracy on the full neuron matrix, with the 83.2% majority baseline as the dashed reference. The two metrics disagree: silhouette hovers around zero in every area, classifier accuracy is at ceiling. The class signal is decodable but is not aligned with the high-variance axes PCA picks.

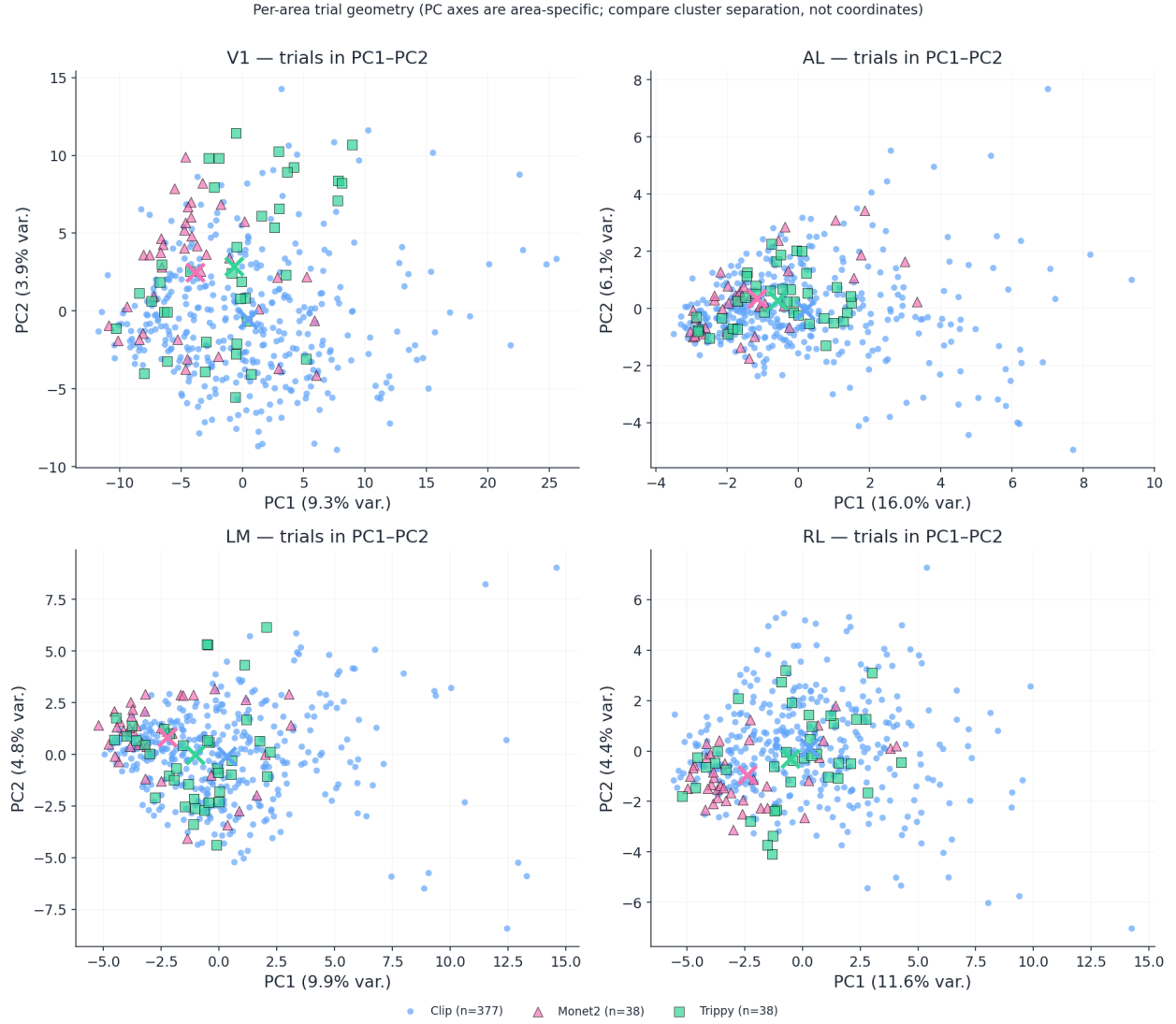


Figure 4: **Trial-averaged PCA per area in PC1–PC2 (Exp. 1).** One panel per cortical area (V1 top-left, AL top-right, LM bottom-left, RL bottom-right). Coloured dots are individual trials; crosses mark class centroids. The three stimulus classes overlap heavily in the high-variance subspace in every area: PCA finds the axes of largest variance, but those are not the ones that separate the classes.

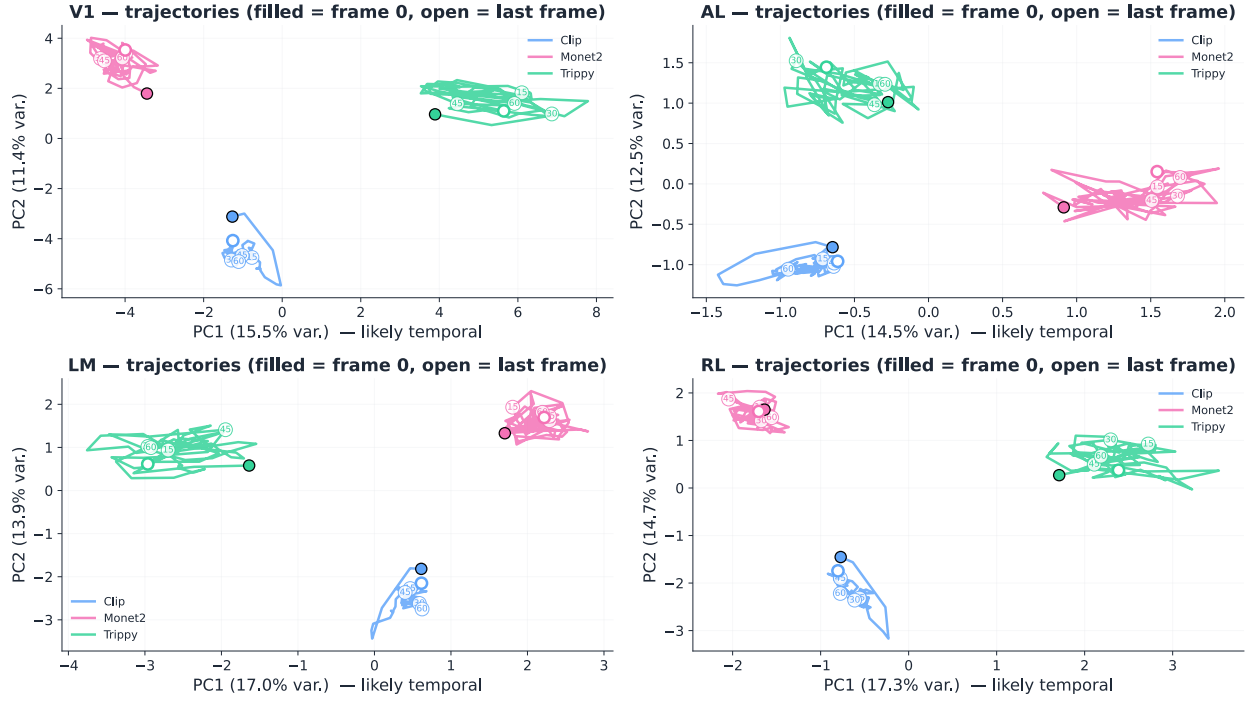


Figure 5: **Trial-averaged trajectories in PC1–PC2 space, per area (Exp. 2).** Four panels (V1, AL, LM, RL): three stationary, non-overlapping clusters per area. The filled circle marks frame 0; the open circle marks the last frame. The frame-0 markers sit offset from each cluster’s bulk, indicating a shared neutral starting point and a rapid jump to stimulus-specific regions of state space within the first few frames (the onset transient seen in Fig. 2).

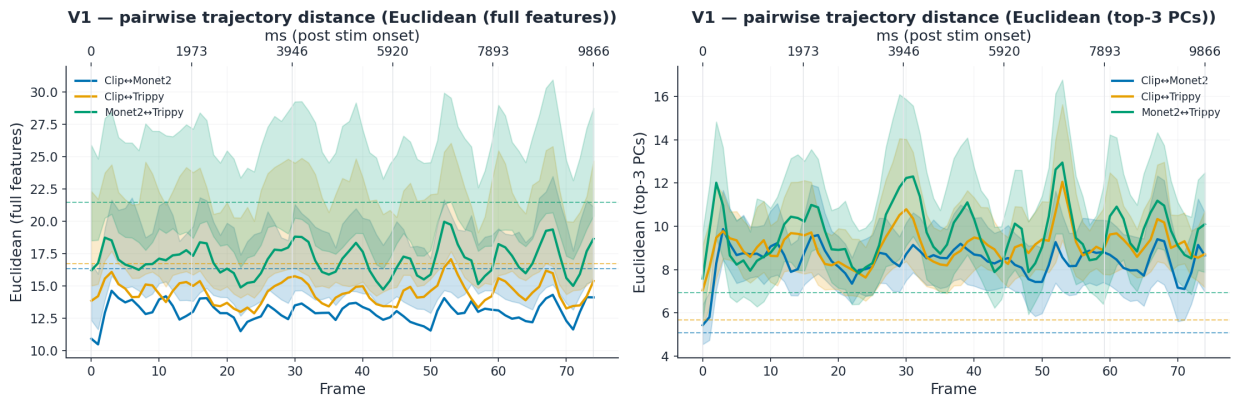


Figure 6: **V1 pairwise distance time courses: biased vs honest (Exp. 2).** Left panel: distances computed in the full neuron space (the honest test). Right panel: distances computed in the top-3 PC subspace fit on real-label trial averages (the biased test). Solid lines are observed pairwise distances per frame; shaded bands are bootstrap envelopes ($B=1000$); dashed horizontals are the 95th percentile of the 100-shuffle null max distance per pair. The biased test puts every pair above its null. The honest test does not: Monet2 $\leftrightarrow$ Trippy stays inside its null band throughout the trial.

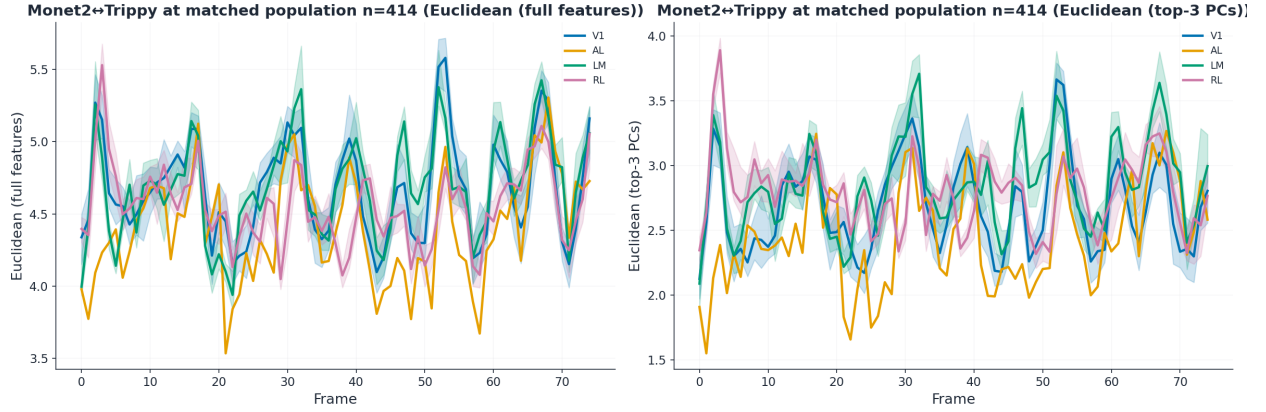


Figure 7: **Population-size control on Monet2↔Trippy (Exp. 2).** Cross-area Monet2↔Trippy distance time course after subsampling V1, LM, RL to AL’s 414 neurons (20 random draws, median $\pm$ IQR shown). Left: full-feature metric. Right: top-3 PC metric. V1’s apparent lead in the raw comparison vanishes once neuron counts are equalised; all four areas converge to the same per-neuron band, confirming that V1’s nominal advantage was a population-size effect rather than a per-neuron one.

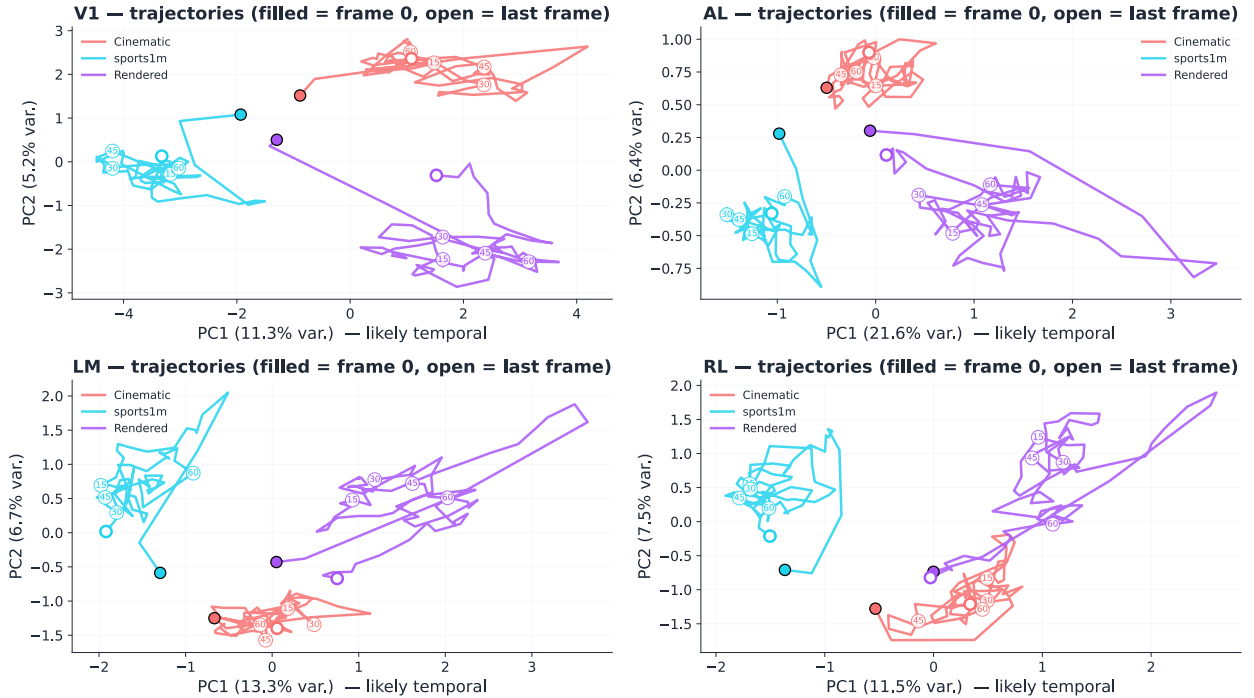


Figure 8: **Clip-category trajectories in PC1–PC2 space, per area (Exp. 3).** Same layout as Fig. 5 but coloured by Clip content sub-category (Cinematic, sports1m, Rendered) rather than by stimulus class. All three categories separate cleanly in every area and remain stationary within their own region of state space after the onset transient.

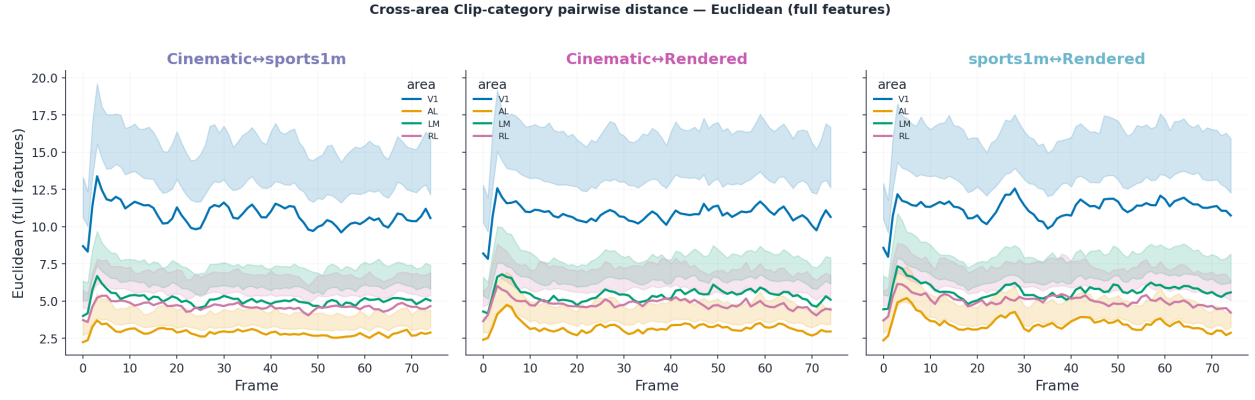


Figure 9: **Cross-area Clip-category distance time courses, all three pairs (Exp. 3).** Full-feature pairwise distance per frame, one panel per category pair (Cinematic↔sports1m, Cinematic↔Rendered, sports1m↔Rendered), with one coloured line per cortical area and bootstrap envelopes shaded. All four areas separate all three pairs at $p \leq 0.02$; onset is at frame 2 (~ 267 ms post stimulus onset).



Figure 10: **Within-Clip sub-category distances after population matching (Exp. 3).** Same three pairs as Fig. 9 but with every area subsampled to AL’s 414 neurons (20 random draws, median $\pm$ IQR). Top row: full-feature metric. Bottom row: top-3 PC metric. AL leads on Cinematic↔Rendered and sports1m↔Rendered (the two natural-vs-synthetic contrasts) but collapses into the other areas on Cinematic↔sports1m (natural-vs-natural). The naturalness axis from Exp. 2 recurs within Clip.